

Answer all questions. Calculators and mobile phones are NOT allowed.

1. [10 pts.] Let $A = \begin{bmatrix} 2 & 3a \\ a & 2 \end{bmatrix}$. Find all values of a , if any, such that $AA^T = A^T A$.
2. [10 pts.] Let A be a 3×3 matrix. Show that if AA^T is skew-symmetric, then $A = O$.

3. [10 + 10 = 20 pts.] Let $(A^T)^{-1} = \begin{bmatrix} 2 & 0 & 0 \\ 2 & -1 & 0 \\ 3 & 4 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 4 & 5 & 6 \end{bmatrix}$.

- (a) Without computing A , find the second row of BA .
 - (b) Find a 3×3 matrix C such that $AC = B$.
4. [10 + 10 = 20 pts.] Given that

$$\begin{array}{rrcr} x_1 & & -x_3 & = & -2a \\ 2x_1 & -x_2 & +2x_3 & = & 0 \\ 3x_1 & +4x_2 & -19x_3 & = & -a. \end{array}$$

- (a) Find all values of a , if any, for which the given system has no solution.
- (b) Solve the given system for $a = 0$.

5. [10 + 10 = 20 pts.] Let $A^{-1} = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 1 \\ 3 & 7 & 0 \end{bmatrix}$.

- (a) Find A .

- (b) Solve the system $AX = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

6. [5 × 4 = 20 pts.] In each of the following questions, select the correct answer.

I. If A is an $n \times n$ nonsingular matrix, then $A^2 \text{col}_j(A^{-1})$ is equal to:

- (A) $\text{col}_j(A)$
- (B) $\text{col}_j(I_n)$
- (C) $\text{col}_j(A^{-1})$
- (D) A
- (E) O .

II. Let A be a $m \times n$ matrix. If the linear system $AX = B$ has no solution, then:

- (A) $m > n$
- (B) $m < n$
- (C) $B = 0$
- (D) A is nonsingular
- (E) $B \neq 0$.

III. Let A , B and C be $n \times n$ matrices. If ABC is nonsingular, then :

- (A) $AB + C$ is nonsingular
- (B) BC is nonsingular
- (C) $BC + A$ is nonsingular
- (D) A is singular
- (E) $A + B$ is nonsingular.

IV. Let A , B and C be $n \times n$ matrices. If $AB = BC = CA$, then $(ABC)^2$ is equal to:

- (A) ABC
 - (B) $(ABC)^3$
 - (C) $A^2C^2B^2$
 - (D) CBA
 - (E) $(AB)^3$.
-

1. [10 pts.] Let $A = \begin{bmatrix} 2 & 3a \\ a & 2 \end{bmatrix}$. Find all values of a , if any, such that $AA^T = A^T A$.

$$\text{We have } AA^T = \begin{bmatrix} 4 + 9a^2 & 8a \\ 8a & 4 + a^2 \end{bmatrix} \text{ and } A^T A = \begin{bmatrix} 4 + a^2 & 8a \\ 8a & 4 + 9a^2 \end{bmatrix}.$$

Therefore, $AA^T = A^T A \Leftrightarrow a = 0$.

2. [10 pts.] Let A be a 3×3 matrix. Show that if AA^T is skew-symmetric, then $A = O$.

Let $A = [a_{ij}]$ and $AA^T = [c_{ij}]$. Since AA^T is skew-symmetric, $c_{ii} = 0$ for all $0 \leq i \leq 3$. But $c_{ii} = a_{i1}^2 + a_{i2}^2 + a_{i3}^2 = 0$ implies that $a_{i1} = a_{i2} = a_{i3} = 0$ for all $0 \leq i \leq 3$. Hence $A = O$.

3. [10 + 10 = 20 pts.] Let $(A^T)^{-1} = \begin{bmatrix} 2 & 0 & 0 \\ 2 & -1 & 0 \\ 3 & 4 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 4 & 5 & 6 \end{bmatrix}$.

- (a) Without computing A , find the second row of BA .

Since $\text{Row}_2(B)$ consists entirely of zeros, $\text{Row}_2(BA) = \text{Row}_2(B)A = [0 \ 0 \ 0]$.

- (b) Find a 3×3 matrix C such that $AC = B$.

$$C = A^{-1}B = \begin{bmatrix} 2 & 2 & 3 \\ 0 & -1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} 14 & 19 & 24 \\ 16 & 20 & 24 \\ 4 & 5 & 6 \end{bmatrix}.$$

4. [10 + 10 = 20 pts.] Given that

$$\begin{array}{rrcr} x_1 & & -x_3 & = & -2a \\ 2x_1 & -x_2 & +2x_3 & = & 0 \\ 3x_1 & +4x_2 & -19x_3 & = & -a. \end{array}$$

- (a) Find all values of a , if any, for which the given system has no solution.

Since $\left[\begin{array}{ccc|c} 1 & 0 & -1 & -2a \\ 2 & -1 & 2 & 0 \\ 3 & 4 & -19 & -a \end{array} \right]$ is row equivalent to $\left[\begin{array}{ccc|c} 1 & 0 & -1 & -2a \\ 0 & 1 & -4 & -4a \\ 0 & 0 & 0 & a \end{array} \right]$, then the given system has no solution for $a \neq 0$.

- (b) Solve the given system for $a = 0$.

The general solution is $X = \begin{bmatrix} t \\ 4t \\ t \end{bmatrix}$, where t is any real number.

5. [10 + 10 = 20 pts.] Let $A^{-1} = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 1 \\ 3 & 7 & 0 \end{bmatrix}$.

(a) $A = (A^{-1})^{-1} = \begin{bmatrix} 7 & 0 & -2 \\ -3 & 0 & 1 \\ -2 & 1 & 0 \end{bmatrix}.$

(b) Solve the system $AX = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

$$X = A^{-1} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 1 \\ 3 & 7 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 13 \\ 17 \end{bmatrix}.$$

6. [$5 \times 4 = 20$ pts.] In each of the following questions, select the correct answer.

I. If A is an $n \times n$ nonsingular matrix, then $A^2 \text{col}_j(A^{-1})$ is equal to:

(A) $\boxed{\text{col}_j(A)}$

(B) $\text{col}_j(I_n)$

(C) $\text{col}_j(A^{-1})$

(D) A

(E) O .

II. Let A be a $m \times n$ matrix. If the linear system $AX = B$ has no solution, then:

(A) $m > n$

(B) $m < n$

(C) $B = 0$

(D) A is nonsingular

(E) $\boxed{B \neq 0}$.

III. Let A , B and C be $n \times n$ matrices. If ABC is nonsingular, then :

(A) $AB + C$ is nonsingular

(B) $\boxed{BC \text{ is nonsingular}}$

(C) $BC + A$ is nonsingular

(D) A is singular

(E) $A + B$ is nonsingular.

IV. Let A , B and C be $n \times n$ matrices. If $AB = BC = CA$, then $(ABC)^2$ is equal to:

(A) ABC

(B) $(ABC)^3$

(C) $A^2 C^2 B^2$

(D) CBA

(E) $\boxed{(AB)^3}$.